Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam
(An Autonomous Institution, Affiliated to Anna University, Chennai)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BE CSE VI Semester	
Course Code: UCS3502	**Course Name: Principles of Machine Learning**
Batch: 2024-28	**Academic Year: 2026-2026(Odd)**
Name of the Faculty	**Dr. R. Sneka Nandhini** **Dr. P. Mirunalini** **Dr. M. Saritha**
Total marks: 40 Marks (Assignment 1 & 2)	**Assignment 1 & 2**

COURSE OUTCOMES

On successful completion of this course, the student will be able to

1. Explain the basic concepts of machine learning (K2)
2. Apply supervised algorithms for different classification problems (K5)
3. Apply the concepts of ensemble learning methods (K3)
4. Apply unsupervised and reinforcement learning techniques to various problems (K5)
5. Apply dimensionality reduction and optimization techniques (K3)
6. Analyze the ethical implications of machine learning models and suggest improvements (K4)

Consider the following problem statements to perform and analyse the ML tasks.

UCS3502 - Principles of Machine Learning

Dataset: https://bigdataieee.org/BigData2026/cup/

- Challenge 01: Explainable Suicide Risk Detection Predict AND explain suicide risk from social media posts.
- Challenge 02: Solar Filament Segmentation Computer vision for solar storm forecasting.
- Challenge 03: FinReason Cup — Financial Reasoning & Audit AI agents for reasoning, hedging & auditing financial data.
- Challenge 04: UCF UrbanTwin Sim2Real LiDAR Train on synthetic. Score on real.
- Challenge 05: CarbonGlobe — Ecosystem Forecasting Emulate 40 years of the planet's carbon cycle.
- Challenge 07: X-Bench — Healthcare AI Agents World's first long-horizon healthcare benchmark — built with 20+ hospitals!

Assignment 1

Sl. No.	**Steps / tasks to be performed**	**Cos, KLs**	**PIs, WK, WP, SDG**	**Marks**
ASSIGNMENT - I				
1	Choose one IEEE Big Data Cup 2026 dataset and investigate the associated real-world challenge.	CO1, K1	1.4.1, 12.3.1, WK2, WK3, WP1, SDG4	-
2	Perform a literature survey to identify existing methods, evaluate their limitations, identify research gaps, and formulate a problem statement with measurable objectives for developing an improved solution.	CO1, K4	2.1.3, 3.2.1, 12.3.1, WK2, WK3, WP1, WP3, SDG4	10
3		CO1,	1.1.1, 4.1.2, 12.2.1,	10

	Identify the type of Machine Learning problem (Classification/Regression/Clustering) and justify your choice.	K3	WK2, WK3, WP1, SDG4	
4	Explain the selected dataset, including its application domain, number of instances, attributes/features, target variable(s), data characteristics, and the real-world challenges associated with the dataset.	CO1, K2	1.4.1, 2.1.3, WK2, WK3, WP1, SDG4	10
5	Perform Exploratory Data Analysis (EDA). Handle missing values, outliers, duplicates, and perform feature engineering/pre-processing. Select at least **two** suitable ML algorithms for the chosen problem.	CO1, K3	1.1.1, 1.4.1, 2.1.3, 4.1.2, 13.3.1, WK2, WP3, SDG4	20
	ASSIGNMENT - II			
6	Train at least three suitable Machine Learning models using the selected dataset. Apply appropriate preprocessing, feature engineering, and mathematical/statistical techniques wherever required. Justify the choice of algorithms, computational techniques, software tools, and implementation methodology using a workflow diagram or pseudocode.	CO2, CO3, CO4, K5	1.1.1, 4.1.2, 5.1.2, 12.2.1, 12.3.1, WK3, WK6, WP1, WP2, WP3, SDG4, SDG9	20
7	Explore different design alternatives by experimenting with various preprocessing techniques, feature selection methods, hyperparameter tuning strategies, and Machine Learning algorithms. Compare these alternatives, justify the final design choice, and critically discuss the strengths and limitations of the selected tools, libraries, and techniques for data modelling, simulation, validation, and performance monitoring.	CO5, K5	3.2.1, 3.2.2, 5.2.1, 5.3.1, WK6, WP2, WP3, SDG9	10
8	Evaluate the trained models using appropriate performance metrics relevant to the problem domain. Compare the results using tables, graphs, and statistical analysis. Interpret the findings, identify the best-performing model based on accuracy, computational efficiency, scalability, and interpretability, and provide suitable inferences supported by experimental evidence.	CO2, K5	4.1.2, 5.3.1, 12.3.1, WK3, WK6, WP2, WP3, SDG4, SDG9	10
9	Discuss the ethical, legal, societal, and engineering considerations associated with the proposed solution, including fairness, privacy, security, and relevant standards or regulations. Identify suitable emerging technologies (such as Explainable AI, Generative AI, Federated Learning, Edge AI, AutoML, Quantum Machine Learning, or Graph Neural Networks) that can improve the proposed solution, justify their adoption, and present the overall findings collaboratively as a team with clear integration of contributions from all members.	CO6, K4	6.1.1, 6.2.1, 7.2.2, 8.3.1, 9.2.1, 13.4.1, 13.4.2, WK8, WP7, SDG9	10

Prepare report with the following sections.

a. Abstract
b. Introduction
    i. Problem Statement
c. Literature Survey
d. Proposed System
    i. System Architecture Diagram
e. Implementation and Experiments
    i. Development Environment
    ii. Dataset Description

iii. Implementation with screenshots (Data Collection, Data Preprocessing, Feature Engineering, ML Model, Comparison of ML Models, Deploy ML model in Cloud using Rapid Miner or AWS or Azure) Include description of each step along with screenshots.

f. Results and Discussions
    i. Impact of the project on human, societal, ethical and sustainable development
g. Conclusion and Future work
h. References

Submission instructions:

There should be only team of 3.

Submit the report in LMS in latex and implementation code with screenshots as a single PDF.